Percentages and Rates

Please work the following questions with those around you.

1. Write each of the following as a percent:
   (a) $\frac{1}{4}$
   (b) 0.02
   (c) 2.35

2. The sales tax in a town is 6.5%. How much will you pay on a $140 purchase? How much will you pay if you have a 20% off coupon? Does is matter if the tax is applied before the coupon, or after? Why or why not?

3. You and your freinds wish to share a large pizza (12 inches in diameter). After placing the order, the manager comes out and explains to you that they are all out of large pizzas, and that it will be a long wait until another is ready, but he can offer you two medium pizzas (8 inches in diameter) now. Should you take the medium pizzas, or wait for the large? (It may be helpful to recall that the area of a circle is given by $A = \pi r^2$, where r is the radius of the circle, also assume that all pizzas are perfect circles).

4. A basketball player scores on 40% of 2-point field goal attempts, and on 30% of 3-point field goal attempts. Find the player's overall field goal percentage.

   (a) What is the average of these two percentages? ______________

   (b) Suppose that the player attempted 200 2-point field goals and 100 3-point field goals. How many 2-point shots did they make? How many 3-point shots? What is their overall field goal average ($\frac{\text{total shots made}}{\text{total shots attempted}}$)

   (c) Do we have enough information to really answer the question as it's stated? Why or why not?

5. Your car can drive 300 miles on 15 gallons of gasoline. How many miles per gallon does your car get?

   (a) If you have the same car and it has 12 gallons of gasoline in it's tank, how far can you drive?

6. A map scale indicates that $\frac{1}{2}$ inch on the map corresponds to 3 miles. How many miles apart are two cities which are $2\frac{1}{4}$ inches apart on the map?